SOC Lab Report: Persistence Detection using Splunk

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1 Abstract

This report demonstrates detection of persistence mechanisms in a SOC environment using Splunk. Attackers often establish persistence to maintain access to compromised systems. This scenario simulates registry run keys and scheduled tasks to achieve persistence.

2 Objective

To detect persistence techniques such as registry modifications and scheduled task creation using Splunk.

3 Lab Setup

- Attacker Machine: Kali Linux
- Victim Machine: Windows 11
- SIEM Tool: Splunk (Ubuntu Server)
- Network: Internal Virtual Lab

4 MITRE ATT&CK Mapping

- Tactic: Persistence
- Technique: Registry Run Keys (T1547.001)
- Technique: Scheduled Tasks (T1053)

5 Attack Simulation

Persistence was established using registry run keys and scheduled tasks.

5.1 Commands Used

```
reg add HKCU\Software\Microsoft\Windows\CurrentVersion\Run /v backdoor /t R

schtasks /create /tn "Updater" /tr "cmd.exe" /sc onlogon
```

5.2 Explanation

The registry key ensures execution of a command during login, while the scheduled task runs automatically when the user logs in.

5.3 Screenshot

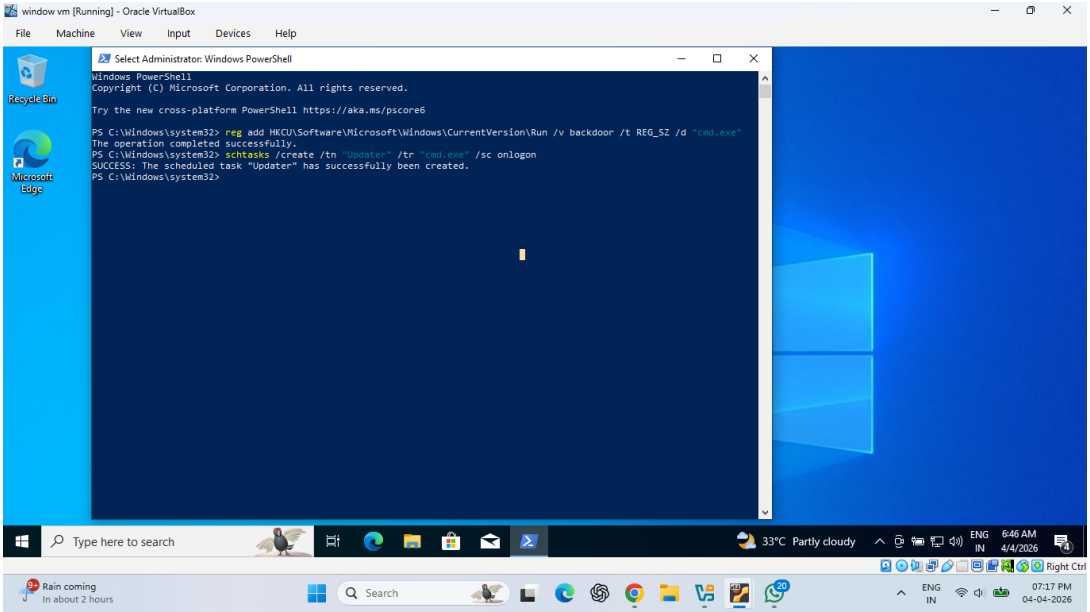

Figure 1: Persistence established using Registry and Scheduled Task

6 Detection in Splunk

Persistence actions are detected through process creation logs (EventCode 4688).

6.1 Detection Query

```
index=windows EventCode=4688
| search Command_Line="*reg add*Run*" OR Command_Line="*schtasks*create*"
| table _time, user, Command_Line
```

6.2 Explanation

The query identifies commands used to create persistence mechanisms.

6.3 Screenshot

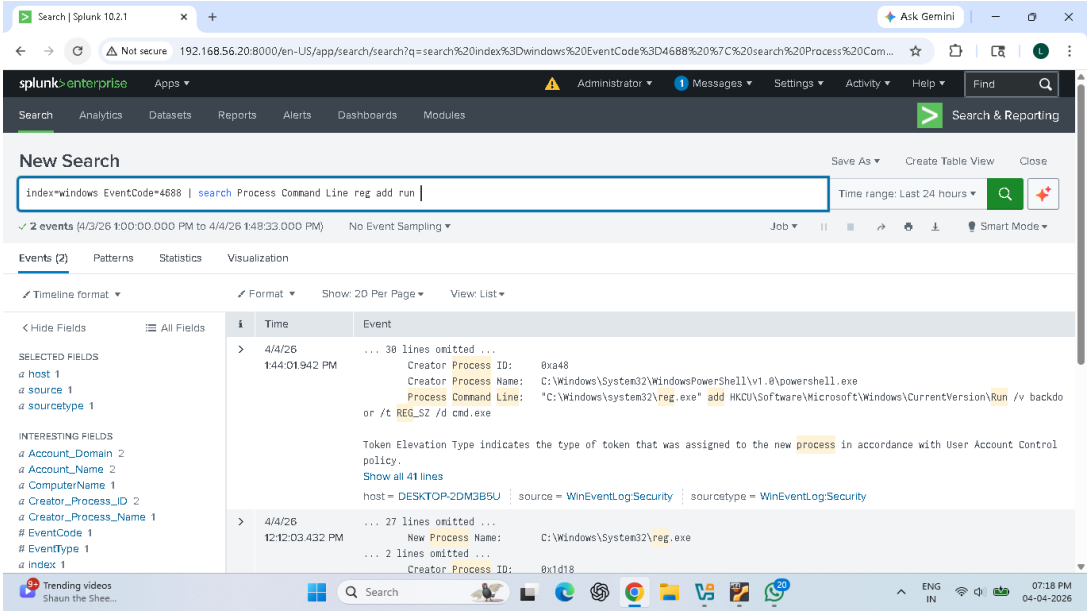

Figure 2: Detection of Persistence Activity in Splunk

6.4 Screenshot

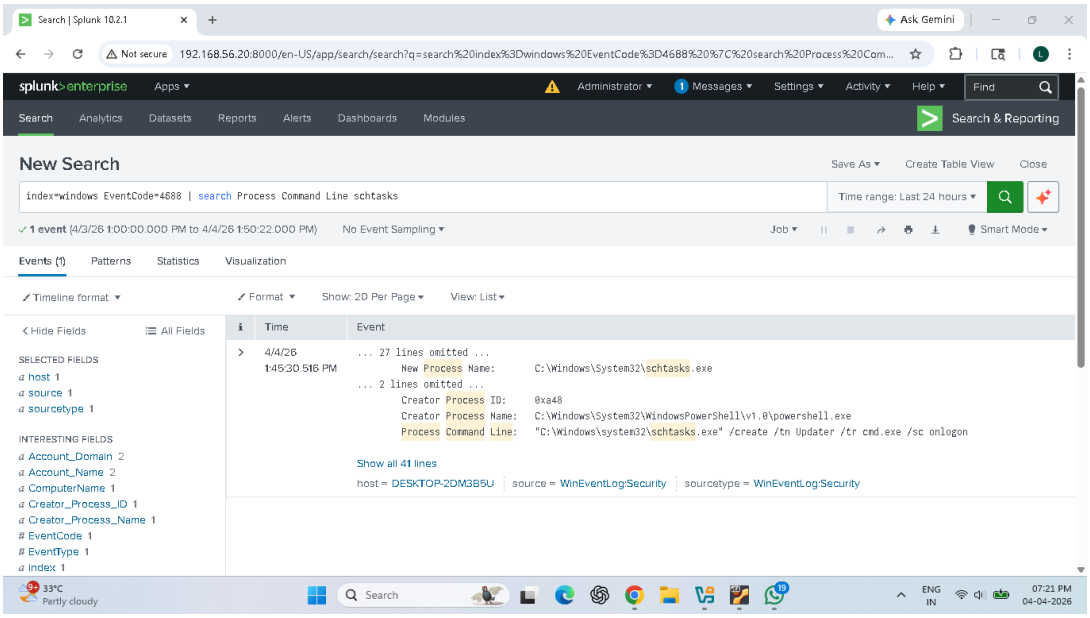

Figure 3: Detection of Persistence Activity in Splunk

7 Investigation

Further analysis was performed to confirm persistence behavior.

7.1 Timeline Query

```
index=windows user="TARGET_USER"
| sort _time
| table _time, EventCode, Command_Line
```

7.2 Findings

- Registry modification detected
- Scheduled task created
- Persistence mechanism established

7.3 Screenshot

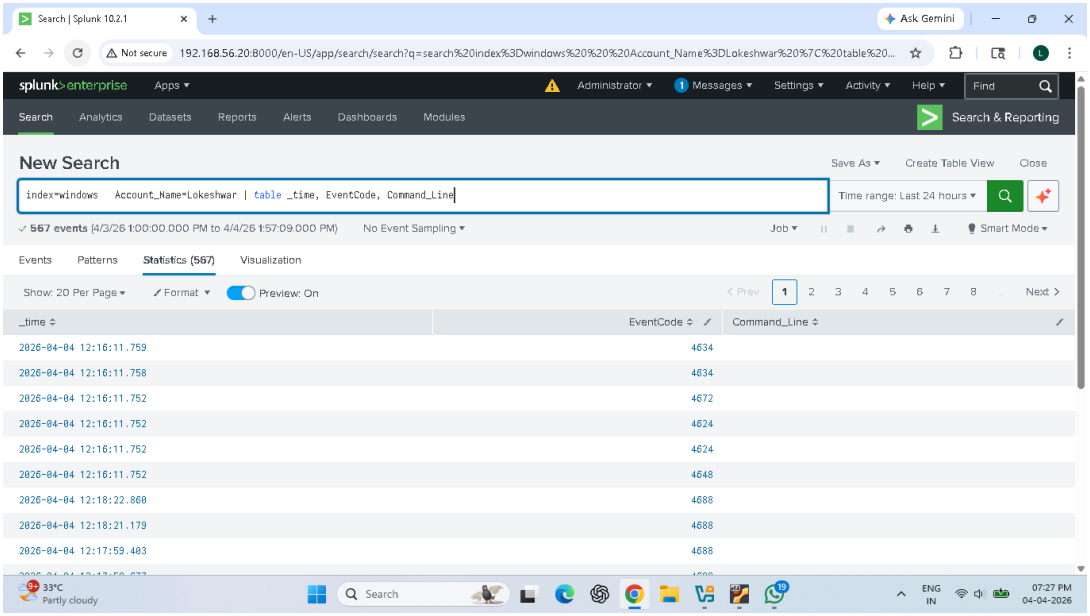

Figure 4: Timeline showing persistence activity

7.4 Screenshot

8 Response Actions

Immediate steps were taken to remove persistence.

- Removed registry run key
- Deleted scheduled task
- Reset user credentials

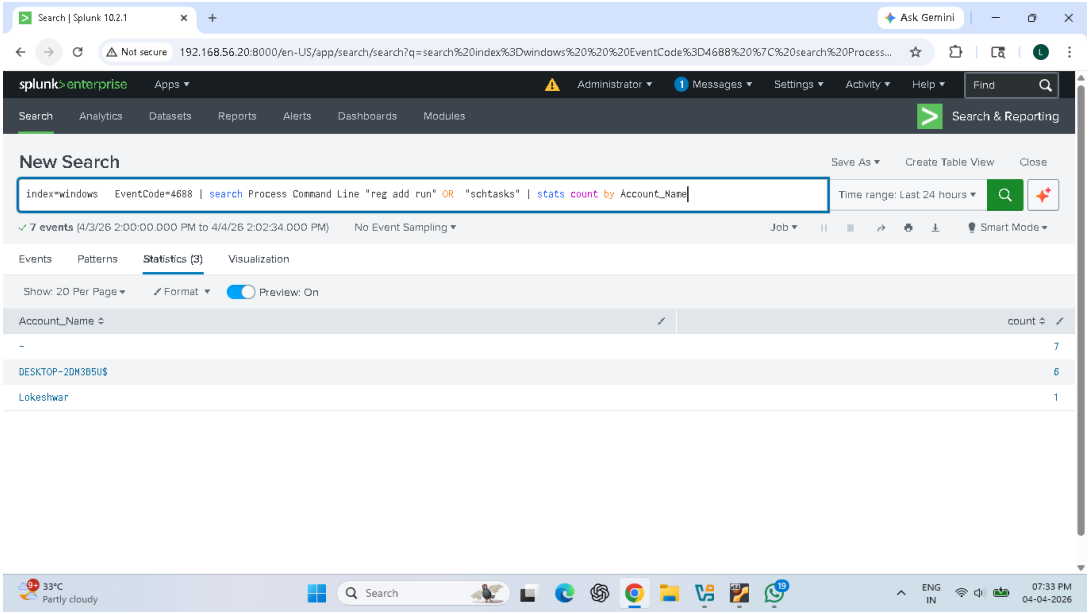

Figure 5: Timeline showing persistence activity

8.1 Commands Used

```
reg delete HKCU\Software\Microsoft\Windows\CurrentVersion\Run /v backdoor /
schtasks /delete /tn "Updater" /f
net user <username> NewPassword@123
```

8.2 Explanation

The persistence mechanisms were removed to prevent automatic execution, and credentials were secured.

8.3 Screenshot

9 Incident Reporting

9.1 Incident Summary

Persistence mechanisms were detected using registry run keys and scheduled tasks.

9.2 Impact

Attackers could maintain access to the system even after reboot.

9.3 Conclusion

The persistence techniques were successfully detected and removed using Splunk and response actions.

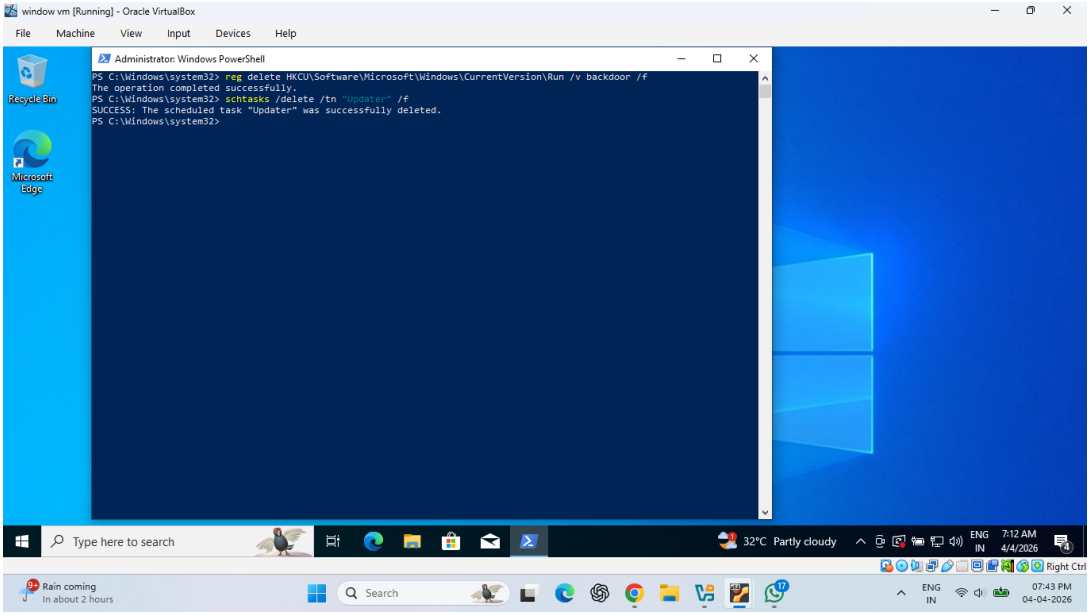

Figure 6: Response Actions - Removing Persistence Mechanisms

10 Conclusion

This scenario highlights the importance of detecting persistence mechanisms to prevent long-term system compromise.